

DA6701

Ex-Post ML techniques for Portfolio Optimization

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1 Problem Setup and Empirical Framework

In this study, we investigate a 12-stock National Stock Exchange (NSE) universe to empirically demonstrate the inherent instability of sample covariance in Markowitz optimization. We aim to quantify the stability and performance gains achieved by applying two distinct regularization techniques: statistical shrinkage (Ledoit–Wolf) and structural clustering (Hierarchical Risk Parity).

Universe ($N = 12$): TCS, INFY, HCLTECH, TATASTEEL, HINDALCO, JINDALSTEL, HDFCBANK, ICICIBANK, KOTAKBANK, HINDUNILVR, ITC, NESTLEIND.

Data Partitioning: Our dataset spans 1,237 trading days (March 2021 to February 2026).

- **Training Horizon:** April 2021 – February 2025 (47 Months / 984 Days).
- **Testing Horizon:** March 2025 – February 2026 (12 Months / 252 Days).

2 Part 1: Statistical Regularization via Ledoit–Wolf Shrinkage

2.1 The “Error Maximizer” and Ill-Conditioning

Standard Mean-Variance optimization acts as an “error maximizer” because it relies heavily on the inversion of the sample covariance matrix (S^{-1}). When S is ill-conditioned, minute estimation errors in asset returns are amplified, leading to extreme and erratic portfolio weights.

We mathematically quantified this instability using the condition number, $\kappa(S) = \frac{\lambda_{\max}(S)}{\lambda_{\min}(S)}$. In our annualized training sample, we observed $\lambda_1 = \mathbf{0.3248}$ and $\lambda_n = \mathbf{0.0091}$, yielding $\kappa_S \approx \mathbf{35.60}$. To prove this instability is structural rather than a finite-sample anomaly, we conducted a bootstrap analysis (300 resamples of 47 months), revealing a mean condition number of **82.2** with variance peaking at **222.9**.

Metric	Mean	Std Dev	Min	Max
Bootstrap Condition Number (κ)	82.2	31.7	31.7	222.9

Table 1: Bootstrap sensitivity analysis demonstrating the extreme instability of the sample covariance matrix across 300 simulated data draws.

2.2 Mathematical Implementation of Ledoit–Wolf

To mitigate this ill-conditioning, we implemented the Ledoit–Wolf (LW) shrinkage estimator. The optimal shrunk matrix Σ_{LW} is derived by minimizing the expected Frobenius norm between the estimator and the true covariance matrix Σ :

$$\min_{\alpha} \mathbb{E} [\|\Sigma_{LW} - \Sigma\|_F^2] \quad \text{s.t.} \quad \Sigma_{LW} = (1 - \alpha)S + \alpha F$$

Here, $F = \mu I$ is the highly structured target matrix scaled by the mean sample variance. The optimal shrinkage intensity α is defined by the ratio of the estimation error of the sample matrix (β^2) to the misalignment between the sample and target (δ^2):

$$\alpha = \max \left(0, \min \left(1, \frac{\beta^2}{\delta^2} \right) \right)$$

2.3 Analytical Results of Spectral Compression

Our algorithmic derivation computed an optimal **shrinkage intensity of $\alpha = 0.1698$ (16.98%)**. Analytically, this means the sample eigenvalues are biased (the largest are historically over-estimated, and the smallest are under-estimated). By applying this α , we artificially compressed the maximum eigenvalue down to **0.2749** and lifted the minimum eigenvalue up to **0.0183**.

Consequently, the condition number dropped to $\kappa_{\text{LW}} \approx 14.96$, achieving a **2.38 $\times$ improvement** in matrix conditioning. As detailed in Table 2, this statistical regularization significantly decoupled the optimizer’s reliance on statistical noise, reducing the Herfindahl-Hirschman Index (HHI) from **0.2792** to **0.2672**.

Asset	Sample MPT	LW-MPT	Performance Metric	Sample MPT	LW-MPT
TCS	20.39%	21.47%	HHI (Concentration)	0.2792	0.2672
HINDALCO	34.39%	31.33%	Ann. Return	22.55%	22.79%
HDFCBANK	31.87%	31.20%	Ann. Volatility	13.70%	13.64%
ICICIBANK	13.35%	16.01%	Sharpe Ratio	1.645	1.671
<i>Others (8 assets)</i>	0.00%	0.00%			

Table 2: Comparison of Maximum Sharpe allocations. Unregularized MPT artificially forces 8 assets to a 0.00% weight due to localized overfitting, a mathematical pathology corrected by LW-shrinkage.

3 Part 2: Structural Regularization via Hierarchical Risk Parity (HRP)

While LW shrinkage resolves matrix inversion errors, MPT still suffers from return-estimation bias. Hierarchical Risk Parity (HRP) addresses this by eschewing returns entirely and structurally regularizing the portfolio using unsupervised machine learning.

3.1 Data-Driven Cluster Discovery

We transformed the correlation matrix ρ into a standardized distance metric:

$$D_{ij} = \sqrt{\frac{1}{2}(1 - \rho_{ij})}$$

Utilizing Ward’s linkage on this distance matrix, our HRP algorithm organically discovered the underlying topology of risk, revealing four statistically robust sub-portfolios entirely separated from rigid GICS industry definitions:

- **Cluster 1 (High Vol/High Ret):** HCLTECH, ICICIBANK, ITC ($\bar{\sigma} \approx 35.4\%$).
- **Cluster 2 (Mod Vol/High Corr):** TCS, JINDALSTEL, NESTLEIND.
- **Cluster 3 (Mod Vol/Low Corr):** INFY, HINDALCO, KOTAKBANK.

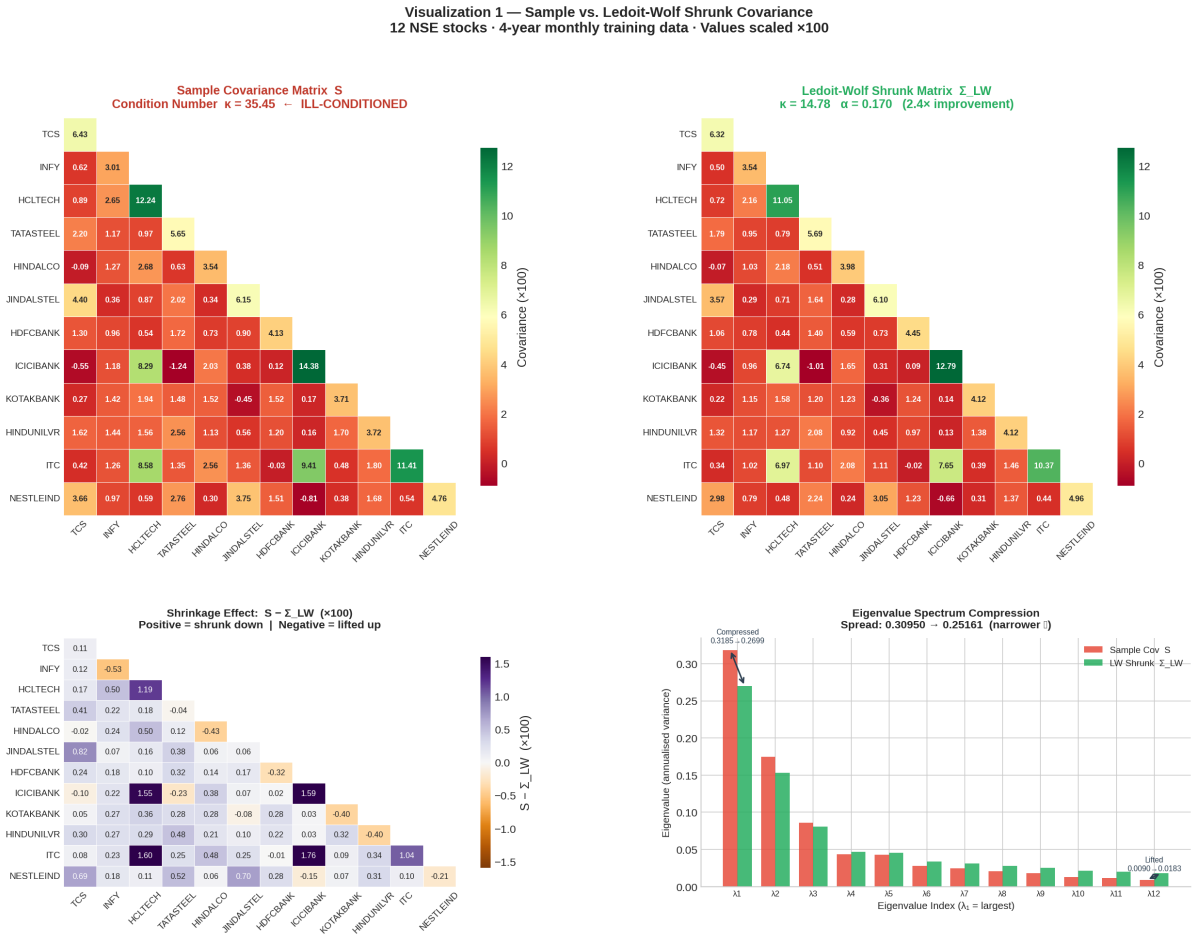


Figure 1: Visualization 1: Covariance Heatmaps and Spectral Compression. The shrinkage effect visibly dampens extreme covariances toward the mean target, physically forcing the ill-conditioned matrix into an invertible state.

- **Cluster 4 (Low Vol/Low Corr):** TATASTEEL, HDFCBANK, HINDUNILVR ($\bar{\sigma} \approx 19.8\%$).

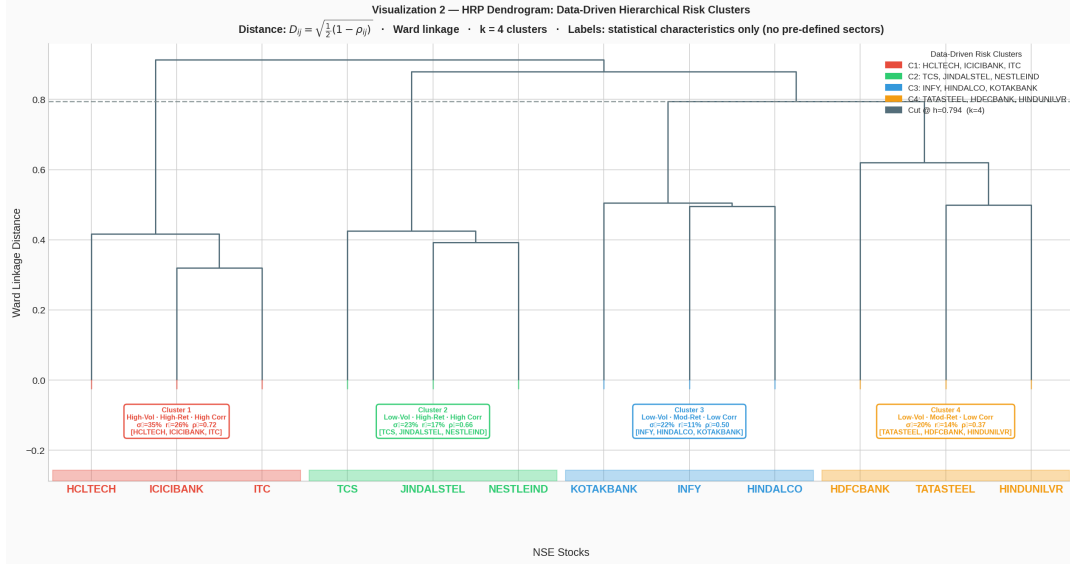


Figure 2: Visualization 2: HRP Dendrogram. The graph maps the empirical behavior of the assets using Ward linkage. Cutting the tree at $h = 0.795$ yields four data-driven risk clusters. **Cluster 1** (Red): High-Vol/High-Ret ($\bar{\sigma} \approx 35\%$, $\bar{r} \approx 26\%$). **Cluster 2** (Green): Low-Vol/High-Ret ($\bar{\sigma} \approx 23\%$, $\bar{r} \approx 16\%$). **Cluster 3** (Blue): Low-Vol/Mod-Ret ($\bar{\sigma} \approx 22\%$, $\bar{r} \approx 10\%$). **Cluster 4** (Orange): Low-Vol/Mod-Ret ($\bar{\sigma} \approx 20\%$, $\bar{r} \approx 13\%$).

3.2 Mathematical Formulation of Recursive Bisection

Following Quasi-Diagonalization of the covariance matrix, we implemented recursive bisection. For any given cluster split into two sub-clusters, L and R , the variance of each sub-cluster is approximated as:

$$V_L = w_L^T \Sigma_L w_L \quad \text{and} \quad V_R = w_R^T \Sigma_R w_R$$

Capital is then allocated inversely proportional to cluster variance, severing the cascading errors typical of full-matrix inversion:

$$\alpha_L = \frac{V_R}{V_L + V_R}; \quad \alpha_R = 1 - \alpha_L$$

3.3 Empirical Weight Allocations

This structural hierarchy naturally prevents corner solutions (where certain assets get a 0% allocation). As shown in Table 3, while standard Min-Variance MPT entirely drops assets like HCLTECH and ITC, HRP successfully distributes capital across all topological clusters, reducing the overall portfolio concentration (HHI) from 0.1484 down to 0.1023.

3.4 Analytical Results of Turnover Friction

A critical metric for the real-world viability of any quantitative strategy is portfolio turnover. Over a 252-day lookback window with monthly rebalancing, standard Min-Variance MPT suffered a highly inefficient average one-way turnover of **15.46%**.

Statistical shrinkage (**LW-MPT**) provided a moderate improvement, reducing average turnover to

Asset	Min-Var MPT	HRP	Equal Wt	Asset	Min-Var MPT	HRP	Equal Wt
TCS	4.84%	9.99%	8.33%	HDFCBANK	16.51%	16.06%	8.33%
INFY	10.69%	8.31%	8.33%	ICICIBANK	1.20%	2.10%	8.33%
HCLTECH	0.00%	4.39%	8.33%	TATASTEEL	9.99%	10.68%	8.33%
HINDALCO	10.66%	8.23%	8.33%	KOTAKBANK	5.78%	10.91%	8.33%
JINDALSTEL	0.69%	5.85%	8.33%	HINDUNILVR	24.27%	13.11%	8.33%
ITC	0.00%	2.72%	8.33%	NESTLEIND	15.35%	7.66%	8.33%
HHI (Concentration Penalty)				Min-Var MPT: 0.1484 HRP: 0.1023			

Table 3: Comparative daily return allocations. HRP completely eliminates the localized 0% weight dropouts seen in standard Min-Variance MPT, ensuring structural diversification.

13.75% (an **11.1% reduction**). However, because **HRP** allocates capital via the localized tree structure mathematically defined above, a change in one stock’s variance only affects its immediate neighbors in the dendrogram, preventing global weight cascading. As a result, HRP achieved an average one-way turnover of just **9.11%**, representing a massive **41.1% reduction** in rebalancing friction.

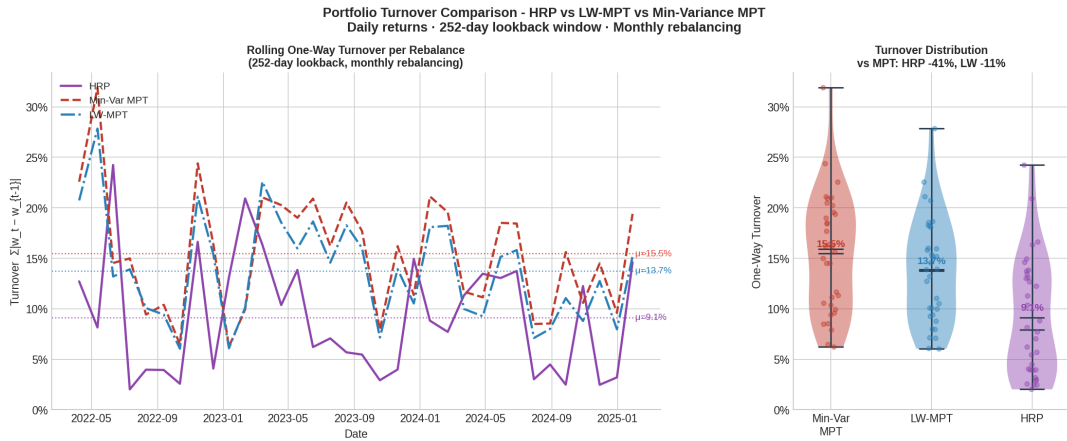


Figure 3: Visualization 3: Portfolio Turnover Comparison. The violin distributions clearly demonstrate the narrower spread and significantly lower median turnover achieved by structurally regularized models (-41%).

4 Crisis Analysis: The SVB Banking Stress (2023)

To practically demonstrate the mathematical theories above, we tracked the weight sensitivity (σ_w) of banking stocks over a 12-month rolling window during the 2023 SVB banking crisis.

During a crisis, asset correlations mathematically converge toward 1. As correlations spike, the determinant of the standard covariance matrix approaches 0, causing S^{-1} to explode. As seen in [Figure 4](#), standard MPT reacted by violently thrashing its allocations. HDFCBANK’s weight swung wildly ($\sigma_w = 35.52\%$).

Statistical regularization (LW-MPT) buffered this by physically imposing a floor on the eigenvalues, dampening the volatility to **30.83%**. However, structural regularization (HRP) proved vastly superior: because it views HDFCBANK as merely one localized element within a broader cluster, its allocation remained exceptionally stable ($\sigma_w = 4.90\%$).

Banking Asset	Standard MPT (σ_w)	LW-MPT (σ_w)	HRP (σ_w)
HDFCBANK	35.52%	30.83%	4.90%
ICICIBANK	15.84%	11.31%	2.70%
KOTAKBANK	8.15%	8.67%	3.71%

Table 4: Weight volatility summary during the 2023 SVB banking stress period. HRP structural regularization effectively eliminates erratic crisis rebalancing.

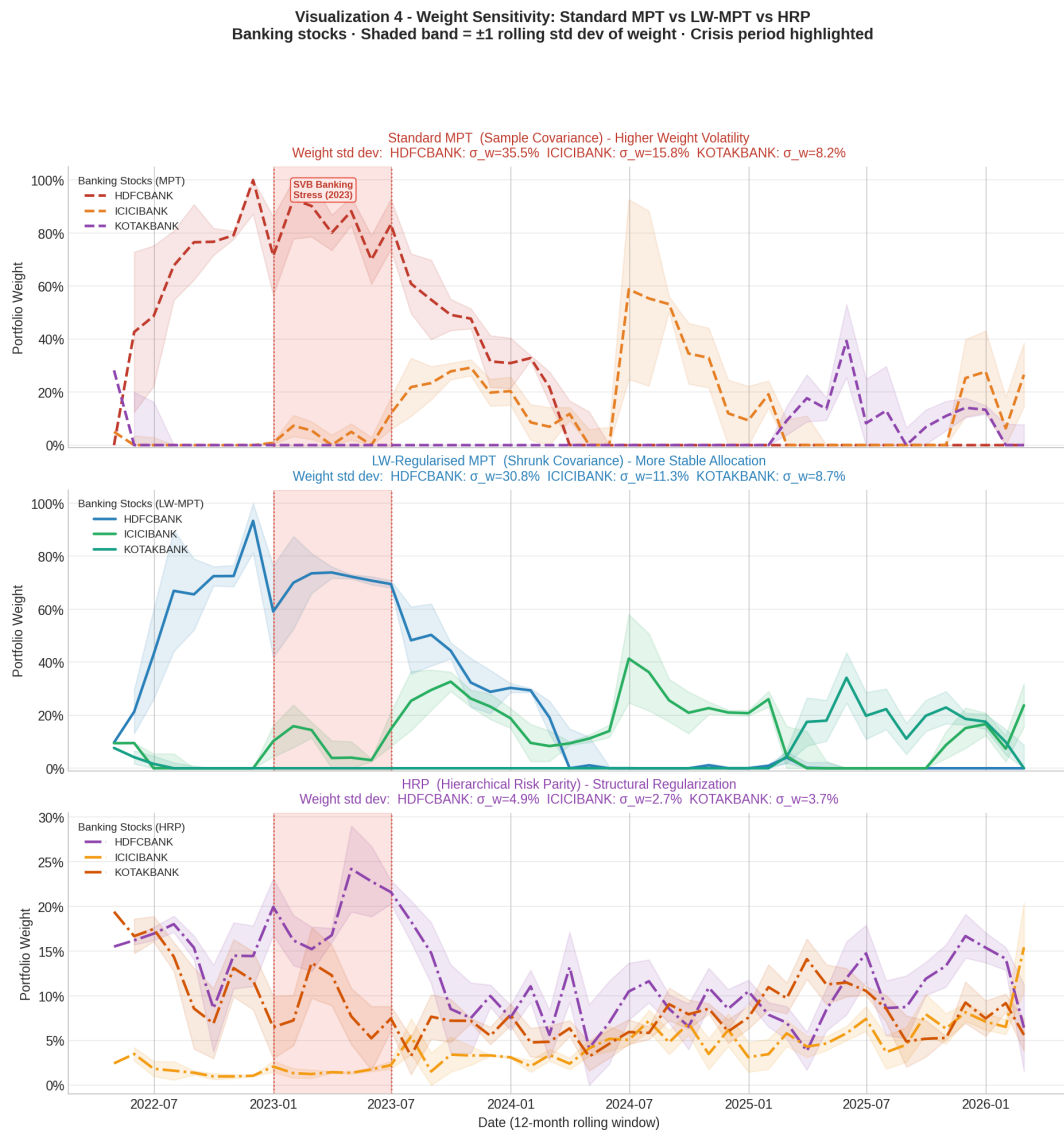


Figure 4: Visualization 4: Weight Sensitivity (σ_w) during Market Stress. The shaded region highlights the 2023 crisis. Standard MPT breaks down into erratic trading, while HRP maintains a stable allocation.

5 Reflection: Regularization as a Prerequisite

We initially viewed standard Markowitz optimization as a mathematically perfect solution from previous assignments that led to high turnovers and noiser plots. We've seen firsthand how relying on an estimated covariance matrix from a limited historical sample introduces massive flaws—specifically, severe ill-conditioning and in-sample overfitting.

The empirical results made it incredibly clear that unconstrained maximizers are essentially "error maximizers." They blindly trust historical noise, taking tiny statistical flukes in our trailing data and blowing them up into heavily concentrated, practically untradable bets.

- **Statistical Shrinkage (Ledoit-Wolf)** It's a necessary step to physically condition the underlying data. It allows us to safely invert the covariance matrix without our portfolio weights exploding, while keeping moderate turnover in check.
- **Structural Machine Learning (HRP)** By bypassing matrix inversion entirely, it lets us build portfolios that drastically minimize rebalancing friction and actually hold up when exogenous shocks hit the market.

Regularization is a prerequisite for real-world quantitative trading.